**Tobias Mwangi-Hartley**
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Dear Quillstone Payments Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. My experience combines Python-based modelling, feature engineering, GCP workflows, scheduled notebook jobs, BigQuery batch scoring, experimentation and automated testing.

At Thistlebox, I developed a weekly recipe-demand forecasting workflow using Python and LightGBM, with a scheduled Vertex AI Workbench notebook job that writes forecast results to BigQuery for operational reporting in Looker. The work improved WAPE from 23% to 14% and was associated with an estimated 9% reduction in food waste. I also built an XGBoost churn model with weekly batch scoring in BigQuery, using dbt for the workflow; a 10% holdout evaluation found churn was 2.1 percentage points lower in the treated group. These projects gave me practical experience in feature engineering, cloud-based data workflows and evaluating models against business outcomes.

I have also built a Streamlit application for planning and analysing sequential A/B tests. I wrote the Streamlit application and Dockerfile, while the data platform team handled its Cloud Run deployment. Since 2025, I have also been adding pytest unit tests to forecasting feature code, with coverage now approximately 60%. I would bring this practical, collaborative approach to Quillstone's AI team, contributing experience in operational modelling, scheduled data workflows, batch scoring, experimentation and testing while developing further in production ML engineering.

Although my experience to date has focused on forecasting, churn modelling and experimentation rather than chat assistants or RAG, I have built and evaluated models used in operational decision-making and worked with data-platform colleagues to turn analytical tools into usable workflows. I am particularly interested in applying that foundation to the engineering discipline required for customer-support AI.

I would welcome the opportunity to discuss how my experience in Python modelling, feature engineering, GCP, scheduled notebook workflows, BigQuery batch scoring, experimentation and pytest testing could contribute to Quillstone's AI team. Thank you for considering my application.

Kind regards,
Tobias Mwangi-Hartley